

Unit 12, Lesson 4: Volume of Right Circular Cylinders – Part 1

Benchmark

MA.7.GR.2.3 Solve mathematical and real-world problems involving volume of right circular cylinders.

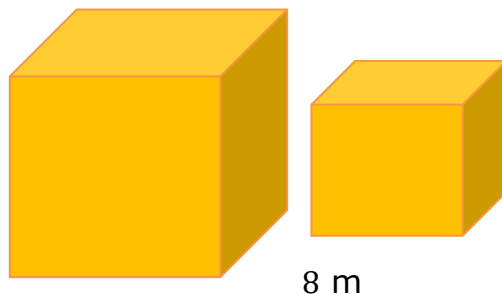
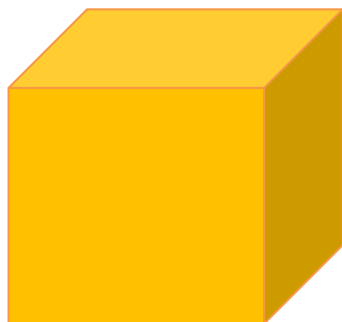
Learning Target

- ☐ I can find the volume of a right circular cylinder.

12.4.1 Warm-Up: Would You Rather?

Gold is worth a significant amount of money. Its value changes with the market. For example, on January 28, 2021 the value of 1 ounce of gold was nearly \$2000.

1. Would you rather have one cube made of solid gold measuring 14 meters (m) on each side, or two cubes made of solid gold, one measuring 12 m on each side and one measuring 8 m on each side?



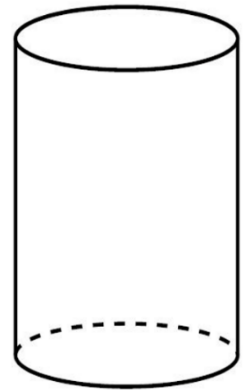
2. Explain your choice.

12.4.2 Activity: Penny Stacks – Volume of a Right Circular Cylinder

1. Stack the given pennies to form a right circular cylinder.

a. How many pennies high is the stack?

b. Compare your stack of pennies with your partner's stack. Suppose both stacks are hollow such that liquid could be poured into them. Discuss with your partner which stack would hold more liquid. Write a summary of your discussion.



Capacity is the amount of space that can be filled in a container. Both capacity and volume are used to measure three-dimensional spaces.

Capacity usually refers to fluid measures, whereas volume is measured in cubic units.

c. Complete the statement.

The capacity, or volume, of the empty cylinder depends on

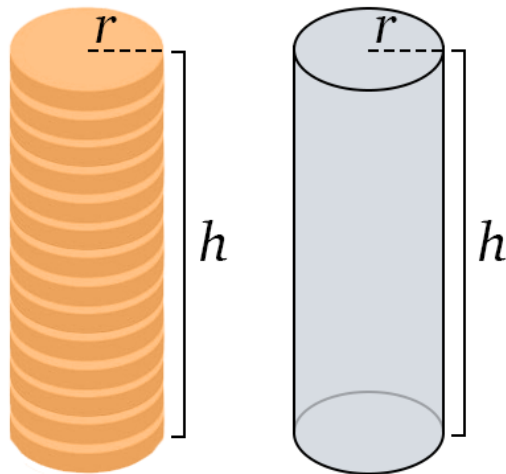
- the radius of each coin.
- the height of the coin stack.
- both the height and radius of the coin

d. Circle the penny stack that has the greatest volume.



12.4.3 Exploration: Discovering the Volume of a Right Circular Cylinder

1. A penny stack with radius, r , and height, h , and an empty cylinder with the same dimensions are shown.

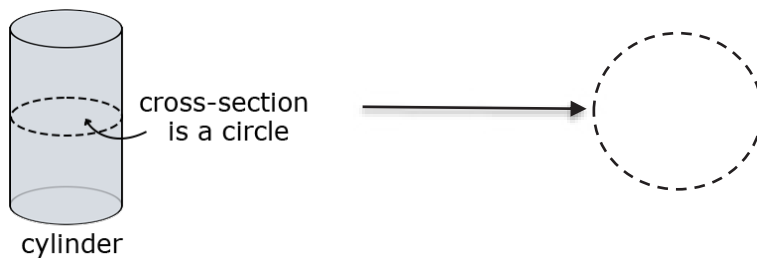


Complete the statement.

The volume of the penny stack is
the volume of the empty cylinder.

- ☐ equal to
- ☐ not equal to

2. A cylinder can be sliced to create a cross section.



Complete the statements.

The cross section of the cylinder shown is the same size

as the

- ☐ circular base
- ☐ lateral area
- ☐ height

of the cylinder.

The expression

- ☐ πr^2
- ☐ $2\pi r$

represents the area of the cross section of the cylinder.

3. With your partner, discuss the relationship between one of the cross sections of the cylinder and the capacity of the entire cylinder.

a. Summarize your discussion.

b. Complete the statements.

Area measures the two-dimensional space an object

takes up. The

- ☐ radius
- ☐ height

of the empty cylinder

makes it a three-dimensional space, as multiple

layers with the same area are stacked upon each other.

To find the **volume of a cylinder**, multiply the

_____ of the base by the _____ of the

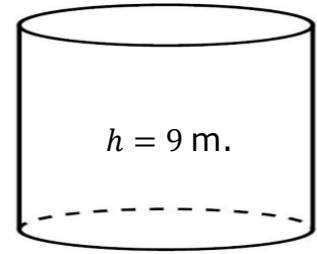
cylinder.

4. Which formula represents the volume of a right cylinder?

- ☐ $V = r \cdot h$, where r is the radius and h is the height.
- ☐ $V = Ch$, where C is the circumference of the base and h is the height.
- ☐ $V = Bh$, where B is the area of the base and h is the height.

12.4.4 Guided Practice: Calculating the Volume of a Right Circular Cylinder

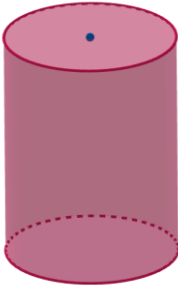
1. A diagram of a right circular cylinder is shown with the radius, r , of 5 meters (m.) and height, h , of 9 m.



- a. Shade in the base of the cylinder.
 - b. Calculate the area of the base. Leave terms of π and use the appropriate units.
 - c. Calculate the volume of the cylinder. Leave in terms of π and use the appropriate units.
2. Consider a cylinder with a radius 5 inches (in.) and a height of 10 in.
- a. Sketch the cylinder and label its dimensions.
 - b. Calculate the volume of the cylinder using $\frac{22}{7}$ to approximate π . Round to the nearest thousandth and use appropriate units.

12.4.5 Your Turn!

1. Suppose three cylinders each have the same height of 8 centimeters (cm.). Cylinder 1 has a radius of 1 cm. Cylinder 2 has a radius of 2 cm. Cylinder 3 has a radius of 3 cm. Label the cylinders with their heights and radii, then find the volume of each cylinder. Use 3.14 to approximate π . Round your answers to the nearest tenth and include appropriate units.

Cylinder Number	Label with Height and Radius	Volume
1		
2		
3		

12.4.6 Wrap-Up: Reflection

1. Complete the table.

The terms and information used in the Exploration and Guided Practice that I knew were ...	Something I wondered during the Exploration and Guided Practice was ...	Something I learned during the Exploration and Guided Practice was ...